

Eighth Section.
Theory of Real Functions.

§ 90. Hermite's solution of Sturm's problem

Another path to the solution of Sturm's problem, leading to simpler results, at least as regards carrying out the calculation in detail, was taken by Hermite; he attaches the question to the law of inertia for quadratic forms and to the Tschirnhausen transformation.¹

Let

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0 \quad (1)$$

be the given equation, and let

$$x_1, x_2, \dots, x_n \quad (2)$$

be its roots, which we assume to be distinct from one another. We use the functions defined in § 68,

$$f_0(x), f_1(x), f_2(x), \dots, f_{n-1}(x), \quad (3)$$

and, as there, put, with $t_0, t_1, \dots, t_{n-1}$ independent variables,

$$y = t_{n-1}f_0(x) + t_{n-2}f_1(x) + \cdots + t_1f_{n-2}(x) + t_0f_{n-1}(x). \quad (4)$$

Let $y_1, y_2, \dots, y_n$ be the values that y assumes for $x = x_1, x_2, \dots, x_n$.

If α is any real value, set

$$H_\alpha = (x_1 - \alpha)y_1^2 + (x_2 - \alpha)y_2^2 + \cdots + (x_n - \alpha)y_n^2. \quad (5)$$

This is a quadratic form in the n variables t , and we first determine the number of its negative and positive terms. If x_1 is a real root, and hence y_1 is real as well, the square $(x_1 - \alpha)y_1^2$ is positive or negative according as x_1 is greater or smaller than α . But if x_1, x_2 form an imaginary pair, then y_1, y_2 are also conjugate imaginary, and

$$(x_1 - \alpha)y_1^2 + (x_2 - \alpha)y_2^2$$

splits into one positive and one negative square. This follows as in § 79 by putting

$$y_1\sqrt{x_1 - \alpha} = u + iv, \quad y_2\sqrt{x_2 - \alpha} = u - iv.$$

It follows that the number N_α of negative squares in H_α is equal to the number of imaginary pairs, increased by the number of real roots that are less than α . If we therefore take a second real number $\beta > \alpha$, form H_β , and denote by N_β the number of its negative squares, then the difference

$$N_\beta - N_\alpha$$

is equal to the number of real roots between α and β .

Thus one obtains a means of determining this number, that is, of solving Sturm's problem, once the coefficients of the function H_α have been represented as functions of the coefficients of $f(x)$ and of α , and the number N_α has then been examined.

¹Hermite, Remarques sur le théorème de M. Sturm. Comptes rendus of the Paris Academy, vol. 36 (1853).

§ 91. Determination of Hermite's form H

To determine Hermite's form H we can use the means learned in the sixth section. In § 72 we derived the formula

$$yf_s = E_{0,s}f_0 + E_{1,s}f_1 + \cdots + E_{n-1,s}f_{n-1}. \quad (1)$$

Using the relations

$$xf_0 = f_1 - a_1, \quad xf_1 = f_2 - a_2, \quad \dots, \quad xf_{n-1} = -a_n,$$

we get

$$xyf_s = E_{0,s}f_1 + E_{1,s}f_2 + \cdots + E_{n-2,s}f_{n-1} - a_1E_{0,s} - a_2E_{1,s} - \cdots - a_nE_{n-1,s}. \quad (2)$$

We therefore determine one more row of functions $E_{-1,s}$ by the equation

$$a_0E_{-1,s} + a_1E_{0,s} + a_2E_{1,s} + \cdots + a_nE_{n-1,s} = 0 \quad (3)$$

and obtain from (1) and (2)

$$\begin{aligned} (x - \alpha)yf_s &= (E_{-1,s} - \alpha E_{0,s})f_0 + (E_{0,s} - \alpha E_{1,s})f_1 + \cdots \\ &\quad + (E_{n-2,s} - \alpha E_{n-1,s})f_{n-1}. \end{aligned} \quad (4)$$

The function $E_{-1,s}$ can, by the relation [§ 72, (7)]

$$a_0t_{n+s} + a_1t_{n+s-1} + \cdots + a_nt_s = 0,$$

be expressed in exactly the same way as the remaining E 's. We therefore obtain the following system of formulae [§ 72, (9)]:

$$\begin{aligned} E_{-1,s} &= a_0t_{n+s} + a_1t_{n+s-1} + \cdots + a_st_n = -a_{s+1}t_{n-1} - \cdots - a_nt_s, \\ E_{0,s} &= a_0t_{n+s-1} + a_1t_{n+s-2} + \cdots + a_st_{n-1} = -a_{s+1}t_{n-2} - \cdots - a_nt_{s-1}, \\ E_{1,s} &= a_0t_{n+s-2} + a_1t_{n+s-3} + \cdots + a_st_{n-2} = -a_{s+1}t_{n-3} - \cdots - a_nt_{s-2}, \\ &\quad \dots \\ E_{s-1,s} &= a_0t_n + a_1t_{n-1} + \cdots + a_st_{n-s} = -a_{s+1}t_{n-s-1} - \cdots - a_nt_0, \\ E_{s,s} &= a_0t_{n-1} + a_1t_{n-2} + \cdots + a_st_{n-s-1}, \\ E_{s+1,s} &= a_0t_{n-2} + a_1t_{n-3} + \cdots + a_st_{n-s-2}, \\ &\quad \dots \\ E_{n-1,s} &= a_0t_s + a_1t_{s-1} + \cdots + a_st_0. \end{aligned} \quad (5)$$

Introduce a second system of variables τ and put

$$z = \tau_{n-1}f_0 + \tau_{n-2}f_1 + \cdots + \tau_0f_{n-1}.$$

Then (4), using the formulae [§ 68, (6)], gives

$$S(f_s) = (n - s)a_s, \quad S[(x - \alpha)yz] = na_0 \sum_{s=0}^{n-1} (E_{-1,s} - \alpha E_{0,s})\tau_{n-s-1} \quad (1)$$

$$+ (n - 1)a_1 \sum_{s=0}^{n-2} (E_{0,s} - \alpha E_{1,s})\tau_{n-s-1} \quad (2)$$

$$+ \cdots + a_{n-1} \sum_{s=0}^0 (E_{n-2,s} - \alpha E_{n-1,s})\tau_{n-s-1}. \quad (6)$$

This passes directly into the function H when $z = y$ is put.

For example, in the case $n = 3$ one obtains

$$\begin{aligned} E_{-1,0} &= -a_1t_2 - a_2t_1 - a_3t_0, & E_{0,0} &= a_0t_2, \\ E_{-1,1} &= -a_2t_2 - a_3t_1, & E_{0,1} &= -a_2t_1 - a_3t_0, \\ E_{-1,2} &= -a_3t_2, & E_{0,2} &= -a_3t_1, \\ E_{1,0} &= a_0t_1, & E_{2,0} &= a_0t_0, \\ E_{1,1} &= a_0t_2 + a_1t_1, & E_{2,1} &= a_0t_1 + a_1t_0, \\ E_{1,2} &= -a_3t_0, & E_{2,2} &= a_0t_2 + a_1t_1 + a_2t_0, \end{aligned}$$

and from these the coefficients of the form H are readily computed from (6):

$$\begin{aligned} H_{2,2} &= -a_0(a_1 + 3a_0\alpha), \\ H_{1,1} &= -3a_0a_3 - a_1a_2 - 2(a_1^2 - a_0a_2)\alpha, \\ H_{0,0} &= -a_2a_3 + (2a_1a_3 - a_2^2)\alpha, \\ H_{1,0} &= -2a_1a_3 + (3a_0a_3 - a_1a_2)\alpha, \\ H_{2,1} &= -2a_0(a_2 + a_1\alpha), \\ H_{0,2} &= -a_0(3a_3 + a_2\alpha). \end{aligned}$$

§ 92. The determinant of Hermite's form

For the question of the number of negative squares of the form H , knowledge of its determinant is important. This determinant can be computed in general as follows.

Put

$$H = \sum_{i,k=0}^{n-1} h_{i,k} t_i t_k.$$

The coefficients $h_{i,k}$ whose determinant is to be formed are given by formula § 91, (6). Thus $h_{i,k}$ is the coefficient of $t_i t_k$ in that formula, namely

$$h_{i,k} = S((x - \alpha)f_{n-i-1}f_{n-k-1}).$$

By the multiplication theorem the determinant formed from these n^2 quantities can be written as

$$\Delta = \begin{vmatrix} (x_1 - \alpha)f_0(x_1) & (x_1 - \alpha)f_1(x_1) & \cdots & (x_1 - \alpha)f_{n-1}(x_1) \\ \vdots & \vdots & & \vdots \\ (x_n - \alpha)f_0(x_n) & (x_n - \alpha)f_1(x_n) & \cdots & (x_n - \alpha)f_{n-1}(x_n) \end{vmatrix} \begin{vmatrix} f_0(x_1) & \cdots & f_0(x_n) \\ f_1(x_1) & \cdots & f_1(x_n) \\ \vdots & & \vdots \\ f_{n-1}(x_1) & \cdots & f_{n-1}(x_n) \end{vmatrix}.$$

Since

$$a_0(x_1 - \alpha)(x_2 - \alpha) \cdots (x_n - \alpha) = (-1)^n f(\alpha)$$

and since the remaining determinant of the $f_i(x_k)$ is a product of two determinants whose square is $a_0^{2n-2}D$, where D denotes the discriminant of $f(x)$ (§ 46), we get

$$\Delta = (-1)^n a_0 f(\alpha) D. \quad (1)$$

Because D is not to vanish, Δ vanishes only when α is put equal to one of the roots of $f(x) = 0$; this case is also to be excluded.

Let a chain of principal minors of Δ , beginning with the lowest, be denoted by

$$\Delta_1(\alpha), \Delta_2(\alpha), \dots, \Delta_n(\alpha).$$

Then by § 83, V, the number of sign changes in

$$1, \Delta_1(\alpha), \Delta_2(\alpha), \dots, \Delta_n(\alpha) \quad (2)$$

is equal to the number N_α of negative squares in H ; and if we count the corresponding number N_β in

$$1, \Delta_1(\beta), \Delta_2(\beta), \dots, \Delta_n(\beta),$$

then $N_\beta - N_\alpha$ is the number of roots of $f(x)$ lying between α and β .

For the cubic equation the expressions are obtained from the end of the preceding paragraph. We form the chain

$$a_0, \quad -H_{2,2}, \quad -\frac{H_{2,2}H_{1,1} - H_{1,2}^2}{a_0}, \quad -\Delta,$$

which plainly has the same number of sign changes as (2), and thereby obtain the four functions

$$\begin{aligned} & a_0, \\ & -3a_0\alpha - a_1, \\ & 2a_2a_0\alpha^2 + \alpha(2a_1a_2 - 9a_0a_3 - a_1^2) - 3a_0a_3 - a_1a_2, \\ & f(\alpha)D. \end{aligned}$$

As a check, take $\alpha = -\infty$, $\beta = +\infty$, and also put $a_0 = 1$. The signs are then determined by

$$\begin{array}{cccc} 1, & +1, & a_1^2 - 3a_2, & D, \\ 1, & -1, & a_1^2 - 3a_2, & -D. \end{array}$$

In counting one must note that, when D is positive, $a_1^2 - 3a_2$ cannot be negative [§ 47, (8)].

§ 93. Elements of the theory of characteristics

Sturm's theorems are generalized in an extraordinary way by Kronecker's theory of characteristics, which pursues for systems of equations with arbitrarily many variables the same aim as Sturm's theorem. We restrict ourselves here to the simplest case, which we shall use for enclosing the complex roots of an equation; and we freely use geometrical intuition and the notation of differential calculus.²

We first consider two real functions $\varphi(x, y)$ and $\psi(x, y)$, and interpret the real variables x, y as rectangular coordinates in a plane. The equations

$$\varphi(x, y) = 0, \quad \psi(x, y) = 0 \tag{1}$$

then represent two curves, which we call simply the curve φ and the curve ψ . We assume first that these curves are closed and do not extend to infinity. We further assume that φ is negative inside the curve φ and positive outside it; similarly, ψ is negative inside ψ and positive outside it.

Let, as Fig. 6 shows, ϑ be the angle made with the positive x -axis by the normal n drawn at any point of φ in the direction in which φ increases, that is, outwards. If $\varphi'(x)$ and $\varphi'(y)$ denote the partial derivatives of φ , and the square root is taken positive, then

$$\cos \vartheta = \frac{\varphi'(x)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}, \quad \sin \vartheta = \frac{\varphi'(y)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}.$$

Draw the tangent t so that it has to this normal n the same position as the positive y -axis has to the positive x -axis, and denote by ξ the angle it makes with the positive x -axis. Then

$$\cos \xi = \frac{-\varphi'(y)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}, \quad \sin \xi = \frac{\varphi'(x)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}.$$

If the advance on φ in the direction t is called positive, and if dx, dy are the projections of this advance, then dx and dy are proportional to, and have the same signs as, $-\varphi'(y)$ and $\varphi'(x)$. If θ is any other function, then

$$d\theta = \theta'(x)dx + \theta'(y)dy$$

²Monatsberichte der Berliner Akademie, March and August 1869, February 1873, February 1878.

has the same sign as the functional determinant

$$[\varphi, \theta] = \varphi'(x)\theta'(y) - \varphi'(y)\theta'(x). \quad (2)$$

With the usual convention for the direction of the coordinates, the positive direction of advance is the one for which the interior of the region lies on the left.

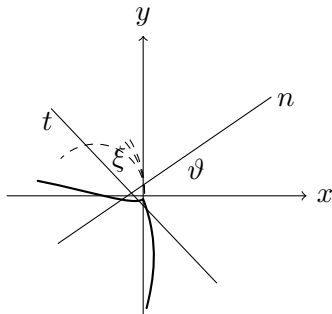


Fig. 6.

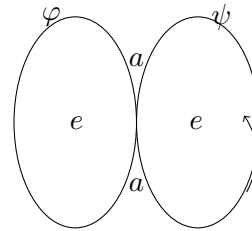


Fig. 7.

If now the positive direction of advance on φ , at an intersection point of the curves φ, ψ , leads into the interior of ψ , then $d\psi$, and hence also $[\varphi, \psi]$, is negative; in the opposite case it is positive (Fig. 7). We call an intersection point of φ and ψ an exit point $A(\varphi, \psi)$ or an entry point $E(\varphi, \psi)$ according as the positive direction on φ leads from the interior of ψ to the exterior, or from the exterior to the interior. Thus

$$\begin{array}{ll} \text{at an } A(\varphi, \psi) & \text{one has } [\varphi, \psi] > 0, \\ \text{at an } E(\varphi, \psi) & \text{one has } [\varphi, \psi] < 0. \end{array} \quad (4)$$

The number of $A(\varphi, \psi)$ is equal to the number of $E(\varphi, \psi)$; and if we interchange φ and ψ , the $A(\varphi, \psi)$ and $E(\varphi, \psi)$ pass into $E(\psi, \varphi)$ and $A(\psi, \varphi)$.

The two functions φ, ψ determine in a unique way a plane region characterized by the fact that in it the product $\varphi\psi$ is negative. We shall call it the inner region (φ, ψ) or (ψ, φ) . In Fig. 7 it is the hatched region.

§ 94. Characteristic of a system of three functions

We now add to the two functions φ, ψ a third function $f(x, y)$. The curve represented by the equation $f = 0$, or the curve f , is also to be closed in the finite part of the plane; and we shall further suppose that it passes through none of the intersection points of φ and ψ .

We give the three functions a definite cyclic order

$$f, \quad \varphi, \quad \psi,$$

so that f follows again after ψ , and with this order we associate a definite, say positive, direction of advance on each of the three curves. We stipulate that with the reversed order

$$f, \quad \psi, \quad \varphi$$

the opposite, i.e. negative, direction of advance is associated on each curve.

In the first order we traverse the curve f in the positive sense and consider the intersection points of f and φ . Such a point is called an exit point $A(f; \varphi, \psi)$ when the positive direction of f leads from the interior of the inner region (φ, ψ) to the exterior, and an entry point $E(f; \varphi, \psi)$ when it leads from the exterior to the interior.

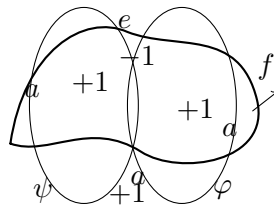


Fig. 8.

In Fig. 8 the points a are exit points and the point e is an entry point. The total number of entry and exit points agrees with the number of intersections of f and φ , and is therefore, since both curves are closed, an even number. If a is the number of points $A(f; \varphi, \psi)$ and e the number of points $E(f; \varphi, \psi)$, then $e - a$ is also even, and

$$\frac{1}{2}(e - a) = k$$

is an integer; following Kronecker it is called the characteristic of the system of functions f, φ, ψ . In the case of Fig. 8 it is equal to -1 .

If we cyclically permute f, φ, ψ , we get three determinations of the characteristic; if we reverse the order and, according to the convention just made, replace the positive directions by the negative ones, we get three more. If the symbols $A(f; \varphi, \psi)$ and $E(f; \varphi, \psi)$ are now also used for the numbers of the corresponding points, these determinations are

1. $\frac{1}{2} [E(f; \varphi, \psi) - A(f; \varphi, \psi)],$
2. $\frac{1}{2} [E(\varphi; \psi, f) - A(\varphi; \psi, f)],$
3. $\frac{1}{2} [E(\psi; f, \varphi) - A(\psi; f, \varphi)],$
4. $\frac{1}{2} [E(f; \psi, \varphi) - A(f; \psi, \varphi)],$
5. $\frac{1}{2} [E(\psi; \varphi, f) - A(\psi; \varphi, f)],$
6. $\frac{1}{2} [E(\varphi; f, \psi) - A(\varphi; f, \psi)].$

The fundamental theorem is that these six determinations give the same number.

To prove it, traverse the curve f in the positive sense and consider all its intersections with φ and ψ . The path along f must enter the inner region (φ, ψ) as often as it leaves it, because it returns to its starting point. But the curve f enters at the points $E(f; \varphi, \psi)$ and $A(f; \psi, \varphi)$ and leaves at the points $A(f; \varphi, \psi)$ and $E(f; \psi, \varphi)$. Thus

$$E(f; \varphi, \psi) + A(f; \psi, \varphi) = A(f; \varphi, \psi) + E(f; \psi, \varphi),$$

which proves the agreement of 1. and 4.

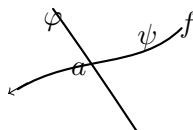


Fig. 9.

Second, if we traverse the curve φ in the negative sense and look at its intersections with f , every point $E(f; \varphi, \psi)$ is also a point $E(\varphi; f, \psi)$, and every point $A(f; \varphi, \psi)$ is also a point $A(\varphi; f, \psi)$. In Fig. 9, where the point a denotes any intersection of f and φ , this point is an $E(f; \varphi, \psi)$ and an $E(\varphi; f, \psi)$ if ψ is positive at a , and an $A(f; \varphi, \psi)$ and an $A(\varphi; f, \psi)$ if ψ is negative at a .

This gives the agreement of 1. with 6. Reapplying the first argument gives the agreement of 6. with 2.; then the second gives that of 2. with 5.; and finally the first gives that of 5. with 3. Hence all six expressions agree. We are therefore entitled to call the number so determined simply the characteristic of the system of functions (f, φ, ψ) .

§ 95. Relation of the characteristic to the intersection points

It must now be observed that the designation of the intersection points of two curves as entry or exit points in § 93 is essentially different from that in § 94. There only the relation of the points to the two intersecting curves was involved, while in § 94 the relation to a third curve also enters. This is expressed completely in the notation $E(\varphi, \psi)$, $A(\varphi, \psi)$ and $E(f; \varphi, \psi)$, $A(f; \varphi, \psi)$. We must now examine more closely how the two modes of designation are related.

For clarity we choose a somewhat simpler figure than before, which at the same time illustrates all possible relations (Fig. 10).

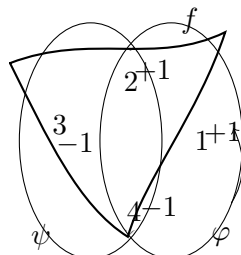


Fig. 10.

At a point $E(\varphi, \psi)$ the curve φ enters the region of the curve ψ ; that is, ψ passes from positive to negative values. If at the same time f is positive, then this point is an $E(\varphi; \psi, f)$, as is point 1 in the figure; if f is negative, it is an $A(\varphi; \psi, f)$, as is point 3. Thus:

A point $E(\varphi, \psi)$ is an	$E(\varphi; \psi, f)$, if $f > 0$,	(point 1 of Fig. 10),
	$A(\varphi; \psi, f)$, if $f < 0$,	(point 3 of Fig. 10),
A point $A(\varphi, \psi)$ is an	$E(\varphi; \psi, f)$, if $f < 0$,	(point 2 of Fig. 10),
	$A(\varphi; \psi, f)$, if $f > 0$,	(point 4 of Fig. 10).

Since at the points $E(\varphi, \psi)$ the functional determinant $[\varphi, \psi]$ is negative, and at $A(\varphi, \psi)$ positive (§ 93), we can summarize the result as follows: $E(\varphi; \psi, f)$ is the number of intersections of φ, ψ at which $[\varphi, \psi]f < 0$, and $A(\varphi; \psi, f)$ is the number of intersections at which $[\varphi, \psi]f > 0$. Hence the characteristic is half the excess of the first number over the second.

This theorem can be expressed in the following way, where it appears wholly independent of magnitudes and dependent only on the relative positions of the curves and their intersections. Distinguish the intersections of φ, ψ according as they are exit points $A(\varphi, \psi)$ or entry points $E(\varphi, \psi)$ by the symbols α and ε ; likewise distinguish them according as they are exterior or interior points with respect to the curve f by a and ϵ . Give the points $\alpha\epsilon$ and εa the character $+1$, and the points αa and $\varepsilon\epsilon$ the character -1 . Then the characteristic of the system (f, φ, ψ) is equal to one half of the sum of the characters of all intersection points of φ, ψ .³

The theorems and definitions just given remain unchanged when the curves under consideration consist of several closed components; double points of the curves may also occur. It is only necessary that for every direction of advance on one of the curves it be completely determined whether that direction is to be considered positive or negative; that is, every piece of a curve f, φ , or ψ must separate regions in which the corresponding function has opposite signs.

Only the cases must be excluded in which one curve passes through a double point of another, or in which the curves touch one another, or in which the three curves f, φ, ψ pass through one point; for in these cases the designation of a point as an entry or exit point would become doubtful. If the curves extend to infinity, they must, in order to make our theorems applicable, be closed by arbitrarily added curve pieces, as we shall see in the next paragraph.

³Kronecker understands something else by the character of a point.

§ 96. Application of characteristics to enclosing the complex roots of an equation

Let now

$$z = x + yi$$

be a complex variable and

$$F(z) = \varphi(x, y) + i\psi(x, y) \quad (1)$$

an integral rational function of z with real or complex coefficients, where $\varphi(x, y)$ and $\psi(x, y)$ are real functions of the real variables x, y . We assume that $F(z)$ and $F'(z)$ do not vanish simultaneously. The function $F(z)$ vanishes only at the intersections of the two curves φ and ψ . Differentiating (1) gives

$$F'(z) = \varphi'(x) + i\psi'(x) = -i\varphi'(y) + \psi'(y),$$

hence

$$\psi'(y) = \varphi'(x), \quad \varphi'(y) = -\psi'(x),$$

and therefore

$$[\varphi, \psi] = \varphi'(x)\psi'(y) - \varphi'(y)\psi'(x) = \varphi'(x)^2 + \varphi'(y)^2.$$

Thus the functional determinant $[\varphi, \psi]$ is never negative and vanishes only where $\varphi'(x)$ and $\varphi'(y)$, and hence also $\psi'(x)$ and $\psi'(y)$, vanish simultaneously.

But this never occurs at an intersection of φ and ψ , for otherwise $F(z)$ and $F'(z)$ would both vanish there. The positive direction of advance is completely determined on every part of the curve φ or ψ by the sign of the function in the adjacent regions; possible double points at which φ , $\varphi'(x)$ and $\varphi'(y)$ vanish simultaneously are no exception.

Since $[\varphi, \psi]$ is positive at all intersections of the curves φ, ψ , all these points are exit points $A(\varphi, \psi)$. It follows that the curves φ and ψ cannot be closed, but must extend to infinity; otherwise an exit point would necessarily be followed by an entry point. Apart from this, the curves φ, ψ here too determine an inner region in which the product $\varphi\psi$ is negative.

We now add to φ, ψ a third function f , whose equation $f = 0$ represents a closed curve, and determine the characteristic of the system exactly as before, by advancing along the curve f :

$$k = \frac{1}{2}[E(f; \varphi, \psi) - A(f; \varphi, \psi)] = \frac{1}{2}[E(f; \psi, \varphi) - A(f; \psi, \varphi)]. \quad (1)$$

Thus, for example, in Fig. 11 the characteristic is $k = 2$.

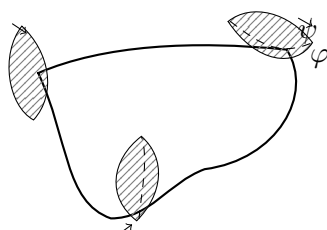


Fig. 11.

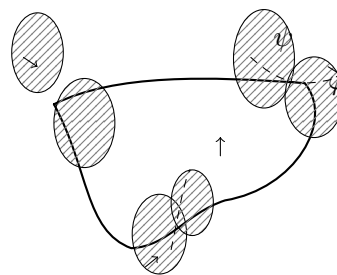


Fig. 12.

To apply our theorems, we must close the curves φ, ψ in some way by arbitrarily drawn connections, all lying outside f . The characteristic of this system is then still determined by (1). As an example, Fig. 12 completes Fig. 11 by drawing such closing connections. The intersections thereby added may be partly exit points $A(\varphi, \psi)$ and partly entry points $E(\varphi, \psi)$. Let their numbers be denoted by

$$A'(\varphi, \psi), \quad E'(\varphi, \psi),$$

while $A(\varphi, \psi)$ denotes the number of originally present intersections. One must always have

$$A(\varphi, \psi) + A'(\varphi, \psi) = E'(\varphi, \psi). \quad (2)$$

We divide the points $A(\varphi, \psi)$ into two groups $A_a(\varphi, \psi)$ and $A_\epsilon(\varphi, \psi)$, the first lying outside and the second inside f . Then the characters of the intersections are

$$\begin{aligned} A_a(\varphi, \psi) &: -1, \\ A_\epsilon(\varphi, \psi) &: +1, \\ A'(\varphi, \psi) &: -1, \\ E'(\varphi, \psi) &: +1. \end{aligned}$$

The double value of the characteristic is therefore, if we omit the notation (φ, ψ) for simplicity (§95),

$$-A_a + A_\epsilon - A' + E' = 2k; \quad (3)$$

adding equation (2) gives

$$k = A_\epsilon. \quad (4)$$

That is, the characteristic is equal to the number of intersections of φ, ψ lying inside f .

We have thus found a means of determining, from the characteristic, the number of roots of the equation

$$F(z) = 0$$

whose geometrical images lie inside an arbitrarily given curve. The arbitrarily added exterior connections play no role in this and may be suppressed entirely.

§ 97. Determination of the characteristic

To make the theorem just proved applicable, we still have to show how to determine characteristics in practice. For this purpose imagine the curve f divided into partial arcs in such a way that in any one of them there lies only one intersection with the curve φ or one intersection with the curve ψ . Let a (Fig. 13) be the beginning, b the end of such an arc, and let ξ be the intersection of f with φ lying in it. Then ξ is a point $E(f; \varphi, \psi)$ if the product $\varphi\psi$ is positive at a , and a point $A(f; \varphi, \psi)$ if $\varphi\psi$ is negative at a ; this gives the possibility of determining the characteristic completely from the signs of φ, ψ at the points dividing the arcs.

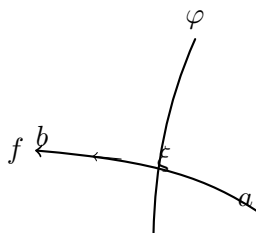


Fig. 13.

How to divide the curve f into such arcs is taught by Sturm's theorem. To this end, think of x and y as functions of a variable t which, along the curve f , runs through all real values from $-\infty$ to $+\infty$ (or also only through the values of a finite interval). Then φ and ψ also become functions of t , and the location of their roots is to be investigated only by Sturm's theorem.

For example, if as in Fig. 14 the curve f is a circle with equation

$$(x - \alpha)^2 + (y - \beta)^2 = \rho^2,$$

then we may put

$$x - \alpha = \rho \frac{1 - t^2}{1 + t^2}, \quad y - \beta = \rho \frac{2t}{1 + t^2},$$

and t simply runs through all values from $-\infty$ to $+\infty$, while the point x, y moves over the whole circle in the positive sense. After multiplication by a suitable power of $1 + t^2$, φ and ψ become integral rational functions of t , to which Sturm's theorem may be applied.

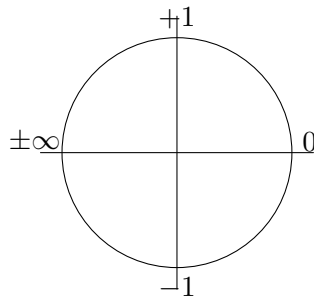


Fig. 14.

§ 98. Gauss's first proof of the fundamental theorem of algebra

Gauss gave three different proofs of the principal theorem of algebra, that an equation of degree n has n roots. The first of them, found in his doctoral dissertation and later further developed and presented differently, rests, although in another guise, on the thought underlying the concept of the characteristic.⁴ We therefore close this section with an account of that proof.

Let $z = x + yi$ and let

$$F(z) = \varphi(x, y) + i\psi(x, y) \quad (1)$$

be an integral rational function of z of degree n , with real or complex coefficients, in which z^n has coefficient 1; thus

$$F(z) = z^n + a_1 z^{n-1} + a_2 z^{n-2} + \cdots + a_n. \quad (2)$$

We again assume that $F(z)$ and $F'(z)$ have no common divisor. The proof that $F(z)$ vanishes for n values of z rests on the fact that, if for f we choose a circle of sufficiently large radius, the characteristic of the system of functions f, φ, ψ can be determined directly and is found to be n . Then § 96 implies that in the chosen circle there are n points z at which $F(z)$ vanishes.

We use polar coordinates and put, since $a_1, a_2, a_3, \dots, a_n$ may be complex,

$$a_1 = p_1 e^{iq_1}, \quad a_2 = p_2 e^{iq_2}, \quad \dots, \quad a_n = p_n e^{iq_n}, \quad z = R e^{i\vartheta},$$

where $p_1, p_2, \dots, p_n$ are positive and the angles $q_1, q_2, \dots, q_n, \vartheta$ may be chosen in any interval of length 2π . Hence

$$\begin{aligned} \varphi(x, y) &= R^n \cos n\vartheta + p_1 R^{n-1} \cos[(n-1)\vartheta + q_1] + \cdots + p_n \cos q_n, \\ \psi(x, y) &= R^n \sin n\vartheta + p_1 R^{n-1} \sin[(n-1)\vartheta + q_1] + \cdots + p_n \sin q_n, \end{aligned} \quad (3)$$

and we also form the derivatives with respect to ϑ :

$$\begin{aligned} \frac{d\varphi}{d\vartheta} &= -nR^n \sin n\vartheta - (n-1)p_1 R^{n-1} \sin[(n-1)\vartheta + q_1] - \cdots - p_{n-1} \sin(\vartheta + q_{n-1}), \\ \frac{d\psi}{d\vartheta} &= nR^n \cos n\vartheta + (n-1)p_1 R^{n-1} \cos[(n-1)\vartheta + q_1] + \cdots + p_{n-1} \cos(\vartheta + q_{n-1}). \end{aligned} \quad (4)$$

We now divide the circumference of the circle of radius R into $4n$ parts, each of angular size

$$2\omega = 2\frac{\pi}{4n},$$

⁴Gauss, *Demonstratio nova theorematum omnium functionum algebraicarum rationalium integram unius variabilis in factores reales primi vel secundi gradus resolvi posse* (1799); Beiträge zur Theorie der algebraischen Gleichungen (1849). Kronecker, Monatsberichte der Berliner Akademie, 21 February 1878.

beginning at $\vartheta = -\omega$. The division points are therefore

$$-\omega, \quad \omega, \quad 3\omega, \quad 5\omega, \dots, (8n-3)\omega,$$

and the intervals are numbered in order as $1, 2, 3, \dots, 4n$. Fig. 15 shows this division for $n = 3$.

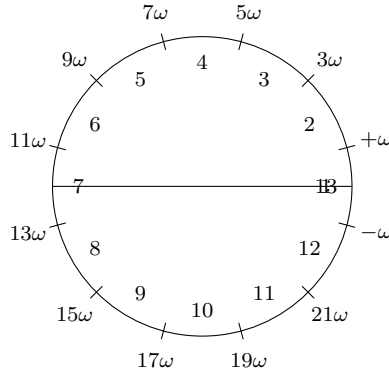


Fig. 15.

In the intervals

$$1, 3, 5, \dots, (4n-1)$$

the absolute value of $\cos n\vartheta$ is greater than $1 : \sqrt{2}$, and its sign is alternately positive and negative. In the intervals

$$2, 4, 6, \dots, 4n$$

the absolute value of $\sin n\vartheta$ is greater than $1 : \sqrt{2}$ and likewise alternates in sign.

But R can be taken so large that in the intervals $1, 3, 5, \dots, (4n-1)$ the signs of φ and $\frac{d\psi}{d\vartheta}$ agree with that of $\cos n\vartheta$, and in the intervals $2, 4, 6, \dots, 4n$ the signs of ψ and $\frac{d\varphi}{d\vartheta}$ agree with that of $\sin n\vartheta$. Since therefore φ changes sign in the intervals $2, 4, 6, \dots, 4n$, it must vanish in each of them; and since the derivative $\frac{d\varphi}{d\vartheta}$ does not change sign there, it can vanish only once in each.

Similarly, ψ must vanish once and only once in each of the intervals $1, 3, 5, \dots, 4n-1$. We have therefore a division of the circumference of the circle of the kind required in the preceding paragraph for determining the characteristic; and since at the division points $\omega, 5\omega, 9\omega, \dots, (4n-3)\omega$ the product $\varphi\psi$ is positive, only points $E(f; \varphi, \psi)$ occur on the circumference, and their number is $2n$. Thus the number of roots of $F(z)$ is equal to n , as was to be proved.